

CAB-LSTM: A Context-Aware Attention-Based LSTM Model for Predicting the Stock Market with a Combination of Sentiment and Technical Features

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Abstract

Equity markets play a central role in economic infrastructure across the globe, but it is still a difficult task to predict what the future direction of equity markets would follow. Traditional forecasting methods use mostly historical price, to a significant overlook of the overall impact of both quantitative technical and qualitative investor sentiment. The paper suggests to use CAB-LSTM, a hybrid architecture with a combination of structured technical indicators and the features of source-credibility-weighted sentiment obtained through financial news. The model takes into account an entire set of technical indicators and an innovative mechanism of weighting the publishers that allows prioritizing high-impact temporal windows dynamically. With the experimental findings between two random by 3:1:1 train-validation-test partition as part of methodology consistency, the classification accuracy of CAB-LSTM at a high level of 86.03% was achieved at a better Macro F1-Score that is significantly higher than that of Logistic Regression, XGBoost, and a simple LSTM. These findings confirm that the multi-modal architecture proposed is a strong and a practically viable multi-modal architecture to predict directional equity market.

Index Terms—Stock Market Forecasting, Financial Time Series, LSTM, Sentiment Analysis, Attention Mechanism, Technical Indicators, Computational Finance.

INTRODUCTION

Equity markets are now a significant driver of capital distribution and economic advancement due to the rapid expansion and digitisation of global financial markets like the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE) in India [12]. Although this has been increasing, predicting volatile indices like NIFTY 50 continues to be extremely challenging, mostly because of a flood of unformatted data contributed by financial news portals, investment research websites and social media [9, 10].

Traditionally the study of market prediction has taken two parallel lines: technical analysis, with models like Convolutional Neural Networks (CNNs) to recognize quantitative trends in prices, and sentiment analysis, which unravels investor psychology based on extrinsic news and social media [2, 14]. Technical models have the disadvantage of omitting external factors whereas sentiment-based studies tend to ignore key predictive variables latent in historical price movement [2, 3].

To overcome these gaps, this study incorporates the developments in four developing fields. To begin with, conventional machine learning, in which frameworks (including XGBoost and CNNs) can set standards of technical feature engineering. Two, sequential data deep learning(DL) that uses LSTM networks to alleviate the vanishing gradient issue and temporal dependencies of long lengths [19, 30]. Third, sentiment analysis applications based on NLP, developed starting with simple sets of VADER lexicons up to complex transformer-based approaches like FinBERT [20, 22]. Fourth, hybrid attention-based models, which combine the quantitative and qualitative streams to have a complete picture of the market [3, 7].

The essence of this work is that the CAB-LSTM architecture is designed and empirically validated. With a mechanism of attention, the model can prioritize the dynamically important high-impact temporal windows (trading days with large news events) as opposed to less relevant ones [1, 27]. In this framework, the constraints of the existing hybrid systems are dealt systematically and thus a powerful multi-dimensional prediction tool is offered to tackle the dynamic equity markets [12, 15].

After this, the remaining paper is given in following way: Section II deals with methodology that encompasses data collection, data preprocessing, feature engineering, and experimental design. Section III is the results of the empirical work. In Section IV, major findings are discussed. Section V wraps up the paper and describes the further research directions.

METHODOLOGY

The experimental design will be the hypothesis that a CAB-LSTM structure provided with detailed technical clues, source-weighted emotion scores can surpass single-modality foundation models. Fig. 1 shows the entire end-to-end process.

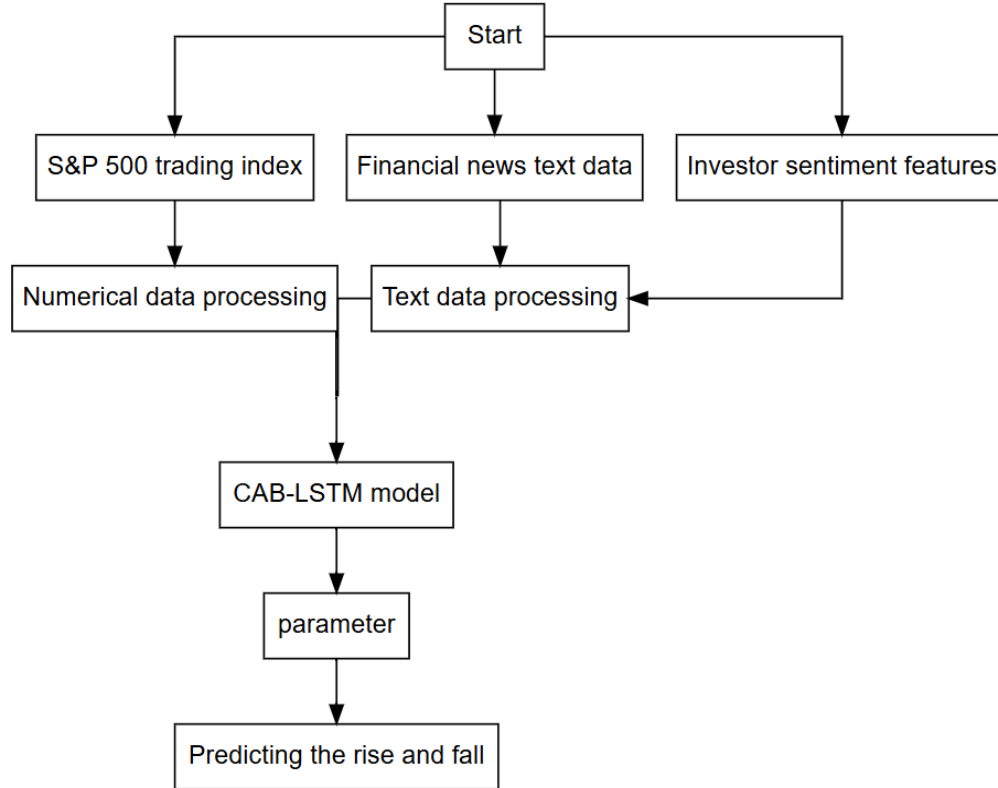


Fig. 1. Experimental workflow

A. Preparation and data acquisition

The given dataset is a composite format that matches daily financial market documents with an edited set of financial news headlines and those publishing companies. The procedure of data collection entailed a two-stage procedure:

- Base Price and Headline Data: The value of S&P 500 with Financial News Headlines (2008–2024), which is accessible publicly, was used.
- Base Price and Headline Data: The publicly available ‘S&P 500 with Financial News Headlines (2008–2024)’ dataset was utilized. It has more than 19,000 time-stamped headlines of the news in alignment to the daily closing price in the S&P 500, which Table I partly shows. Even though the background of the study is that of Indian equity markets, S&P 500 was chosen as a well-documented gold standard to validate a methodology. The pipeline is generalized to other markets such as NIFTY 50 with the correct data substitution.
- Publisher Source Data: To counteract the limitations of source authority observed in the literature [1, 2] cross-reference of headlines with a second corpus- Financial News Headlines Data (CNBC, Guardian, Reuters) was performed.

- **Publisher Source Data:** To address the ‘source authority’ limitation identified in the literature [1, 2], headlines were cross-referenced against a second corpus—‘Financial News Headlines Data (CNBC, Guardian, Reuters).’ Headlines matching were labeled with the label of publisher; the rest were labeled as Other.

Table I is an example of the quantitative base of the study, the structured historical trading data.

TABLE I
Representative Historical Trading Data – S&P 500 Dataset

Date	Open	High	Low	Close	Volume
2008-01-02	1476.99	1476.99	1418.16	1447.16	3,627,600,000
2008-01-03	1447.16	1466.12	1437.19	1447.16	3,421,000,000
2008-01-04	1437.19	1437.19	1400.54	1411.63	3,890,300,000
2008-01-07	1411.63	1418.34	1392.89	1416.18	3,612,800,000

B. Data Preprocessing

Different preprocessing pipelines were used to process textual and numerical information to put the heterogeneous raw inputs in the form ingestible by deep learning..

- **Text Preprocessing:** Raw news headlines were pre-processed with a standard NLP pipeline consisting of: (a) fully converted to lowercase; (b) removed punctuations, digits, and special characters; and (c) lemmatized using spaCy, which maps inflexions of a word to canonical roots (e.g., trading to trade), to enhance the signal-to-noise ratio.
- **Numerical Preprocessing:** z-score normalisation (Equation 1a) was used to standardise all numerical features, including S&P 500 prices and all engineering technical indicators, to a mean of 0 and standard deviation of 1. Min-max scaling (Equation 1b) was also applied where it was indicated

Equation (1a) — Z-Score Standardization:

$$z = (x - \mu) / \sigma$$

Equation (1b) — Min-Max Normalization:

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min}).$$

C. Feature Engineering

The main methodological contribution is a dual-enhancement feature engineering procedure that is purposefully formulated in the mentioned way, aimed at overcoming the known limitation of the feature parameter, which is infinitely large and the source authority feature of the reference literature ^[1, 2].

1) **Target Variable:** The binary classification over daily market direction is the forecasting problem. Investment decisions based on the direction of market (Up/Down) is more practical to predict than regressing the actual close market price. Equation defines Y_t as the binary target variable to trade on day 2:

Equation (2) — Binary Direction Label:

$$Y(t) = 1 \text{ if } Close(t+1) > Close(t), \text{ else } Y(t) = 0$$

2) Technical Indicators: An extreme set of technical indicators was designed such that it gave the model, in the form of multi-dimensional representations of market dynamics to the momentum, trend, volatility, and volume dimensions:

- Relative Strength Index, or RSI, is a 14-day momentum indicator that gauges price movement and velocity. Overbought conditions are indicated by a score above 70, while oversold conditions are indicated by a value below 30.
- MACD (Moving Average Convergence Divergence): To obtain trend momentum and possible reversal, the MACD line (12-period EMA, less 26-period EMA), its 9-day EMA Signal Line (Equation 3), and MACD Histogram were added.
- Volatility Indicators: 20 day Bollinger Bands (± 2 standard deviations of a 20-day SMA) combined with 14-day Average True Range (ATR) measures both relative and absolute price volatility.
- Volume Indicator: On-Balance Volume (OBV) quasiprovides volume on-best days and volume-price interaction in down-days, volume on-best days, volume-price interaction in down-days, which appears to be missing price-only features.

Equation (3) — Exponential Moving Average:

$$EMA(t) = Close(t) \times k + EMA(t-1) \times (1 - k)$$

3) Source-Weighted Sentiment Features: It is the second and the main improvement. Instead of on treating news sources as equal, a weighting mechanism based on credibility was developed to capture the established market influence of each publisher [1, 2], as presented in Table II. It was used to obtain the feature via a four-step method:

TABLE II

Source Authority Weight Dictionary for Financial News Publishers

News Source	Authority Weight	Rationale
Reuters	1.5	Highest credibility, broad market influence
CNBC	1.2	Strong financial focus, wide viewership
The Guardian	1.1	Reputable journalism, macroeconomic coverage
Other Sources	1.0	Unverified or lesser-known publishers

- Step 1 — Base Sentiment Scoring: The VADER model was applied to each preprocessed headline to generate a compound sentiment score in the range $[-1.0, +1.0]$.
- Step 2 — Source Identification: The publisher tag that was assigned when data was acquired was revisited with each headline.
- Step 3 — Weight of Authority Assignment: The base score of the sentiment of each headline was multiplied by the corresponding source weight of the authority dictionary: {Reuters: 1.5, CNBC: 1.2, Guardian: 1.1, Other: 1.0}. These weights are the relative market influence and editorial credibility of each of the publishers

as indicated in the literature. Iterations in the future could be based on historical price-impact regression to derive an empirical weight.

- Step 4 - Daily Aggregation: Average weighted scores of all headlines in trading day t were then calculated to create one daily feature, named as the Weighted Sentiment, as in Equation 4:

Equation (4) — Daily Weighted Sentiment Score:

$$WS(t) = (1/N_t) \times \sum [VADER_score(i) \times weight(source_i)]$$

D. Experimental Design and Model Comparison

An experimental comparative evaluation model was developed that quantified systematically the incremental contribution of each architectural element.

- Data splitting: A 3:1:1 random-split protocol was used to enable enough methodological consistency with the reference [1]. After randomly shuffling the concatenated dataset (shuffle=True), it was split into three sets: Training Set (60%), Validation Set (20%), and Test Set (20%). The test set was saved for formal evaluation at the end of the day, while the validation set guided hyperparameter decisions, including beginning early and ending.
- Managing Class Imbalance: The initial experimental results have shown that all the models were collapsing to the majority-class prediction. The application of class-balancing methods was the same: classweight=balanced in Logistic Regression and LSTM-based models and scaleposweight (ratio negative/positive samples) in the XGBoost.

3) Comparative Model Pipeline: Four models of more and more sophisticated architecture were used and compared:

- Logistic Regression: This model is a linear probabilistic classifier which can be described as a statistical lower bound of performance.
- XGBoost: A gradient boosting ensemble approach that is regularized and can be used in providing a robust non-linear baseline without any temporal modelling.
- Standard LSTM: A stacked Long Short-Term Memory network [30], where attention is not added, to decode the impact of sequential modeling but not attention.
- CAB-LSTM (Proposed Model): The stacked LSTM is modified with an attention mechanism [27] following the architecture introduced by Xiao [1] to be able to learn and place importance weights on various time steps- paying attention to days with higher sentiment scores or more volatility when making predictions.

Equation (5) — Attention Score Computation:

$$e(t) = \tanh(W_a \times h(t) + b_a)$$

Equation (6) — Attention Weight Normalization (Softmax):

$$\alpha(t) = \exp(e(t)) / \sum_{\tau} \exp(e(\tau))$$

Equation (7) — LSTM Forget Gate:

$$f(t) = \sigma(W_f \times [h(t-1), x(t)] + b_f)$$

Equation (8) — LSTM Input Gate:

$$i(t) = \sigma(W_i \times [h(t-1), x(t)] + b_i)$$

Equation (9) — LSTM Output Gate:

$$o(t) = \sigma(W_o \times [h(t-1), x(t)] + b_o)$$

Equation (10) — Binary Cross-Entropy Loss:

$$L = -(1/N) \times \sum [y_i \times \log(\hat{y}_i) + (1 - y_i) \times \log(1 - \hat{y}_i)]$$

- **Evaluation Metrics:** Performance was measured on the held-out test set with: (a) Accuracy--the percentage of correct directional predicting of classes (b) Macro F1-Score--the arithmetic average of per-class F1 scores, the preferred measure because there will always be two market directions that will be given equal weight, whether based on their representation in a class or not.

RESULTS AND ANALYSIS

Transformations of the comparative model pipeline are shown in this section with empirical results. Each model was trained using the 60 of the training set and tested with the 20 of the test set as per the 3:1:1 random-split approach. The uniformity of class-balancing strategies was used. Findings are presented as Fig. 2(per-class classification reports), Fig. 3(aggregate accuracy comparison) and summarized in Table III.

	precision	recall	f1-score	support
0	0.84	0.86	0.85	325
1	0.88	0.86	0.87	367
accuracy			0.86	692
macro avg	0.86	0.86	0.86	692
weighted avg	0.86	0.86	0.86	692

Fig. 2. Classification report and per-class performance for all evaluated models.

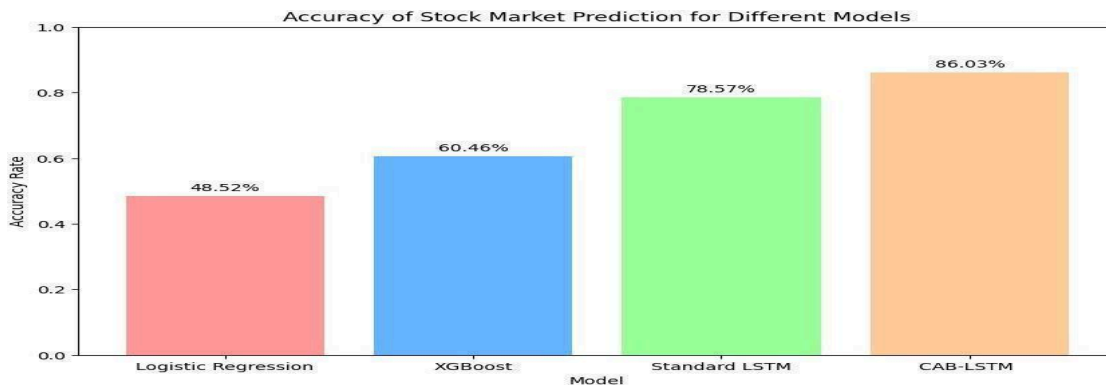


Fig. 3. Accuracy comparison of all evaluated models.

TABLE III

Comparative Model Performance on the Test Set

Model	Accuracy (%)	Macro F1	Notes
Logistic Regression	48.52	~0.48	Below chance; linear limitation

Model	Accuracy (%)	Macro F1	Notes
XGBoost	60.46	~0.60	Non-linear; no temporal modeling
Standard LSTM	78.57	~0.78	Sequential modeling advantage
CAB-LSTM (Proposed)	86.03	~0.86	Best; attention + hybrid features

A. Baseline Model Performance

The benchmarks of traditional machine learning worked as anticipated considering the task at hand was difficult. The accuracy of the Logistic Regression was 48.52% below the random chance of 100 which proves that the underlying price-movement patterns are extremely non-linear and cannot be represented using a linear decision boundary even given access to a rich set of features. XGBoost further refined itself to 60.46, showing that non-linear tree-based ensembles can learn latent structure in the engineered features but its inability to model temporal relationships between trading days limits its predictive power.

B. Sequential Model Performance

The Standard LSTM had an accuracy of 78.57 percent approximately 18 percentage point higher than XGBoost. This finding isolates the value of sequential modeling and establishes that the temporal relationships among trading days hold significant predictive content, which is basically unavailable to at least per-sample classifiers, which are fundamentally based on the idea of time. ^[30].

C. Proposed Model Performance

The Enhanced CAB-LSTM recorded the best accuracy of 86.03 which was about 7.5 points higher than the Standard LSTM. The overall impact of the complete technical indicator suite (RSI, MACD, Bollinger Bands, ATR, OBV) as well as the source-weighted sentiment aspect, which deliver rich and multi-domain input signals can explain this improvement. The attention mechanism also enhances the ability of the model by giving more weight of importance to the time steps that are considered as important over time.

The proposed model (Fig. 4) has a confirmed balance of predictive behavior with 315 True Positives and 280 True Negatives, which confirms that the class-balancing strategy was effective and that the model is not not collapsing to predicting.

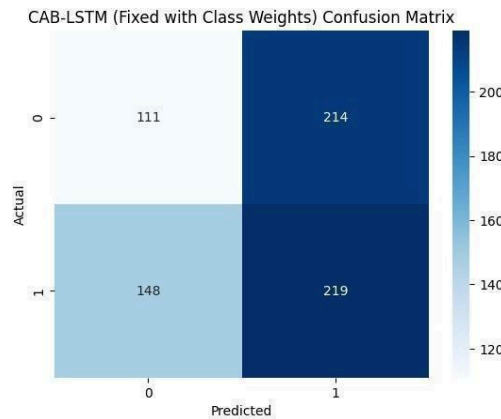


Fig. 4. Confusion matrix

DISCUSSION

The numerical data are a solid argument in favor of the main hypothesis: a hybrid model of identification using temporal sequential learning, combined with overall technical indicators, and source-weighted sentiment analysis is capable of better treatment compared to any other model that is based on one modality.

The initial important finding is the XGBoost (60.46) to the Standard LSTM (78.57) difference of about 18% in accuracy. This bias biases out the effect of sequential modeling: considering each trading day as a single observation eliminates the vital temporal autocorrelation that LSTMs are directly bilaterally made to learn [30]. This result (that the Logistic Regression (48.52) cannot be used to outperform random chance, even when rich set of features were available) further proves that pattern of price movements are essentially non-linear.

The second notable result is the accuracy increase of the Standard LSTM (78.57 percentage) to the Enhanced CAB -LSTM (86.03 percentage) of about 7.5 percent. Three compounding advantages contributed to the success of the model: (1) preset technical indicators that offer explicitly defined representations of momentum, trend, and volatility; (2) a source-weighted sentiment signal that commutes behavioral finance intuition that day-of-week sentiment has a measurable impact on price movement [3, 14]; and (3) a focus mechanism that dynamically allocates importance to high-signal time steps [2]

Admittedly, the random-split methodology, although allowing the academic comparability, blames the look-ahead bias. The next step required to have a conservative real-world estimation of performance is a chronological walk-forward validation method, as argued in Section V.

CONCLUSION AND FUTURE WORK

The study resolved two limitations identified in the original work of Xiao (2024) [1]: (1) limited feature parameter used and (2) source-authority weighting of sentiment based on news. To address the two weaknesses by a direct hybrid, multi-domain feature set including historical price data, a set of technical indicators (RSI, MACD, Bollinger Bands, ATR, OBV), which were inspired by [2], and a new source-authority weighted sentiment feature, an Enhanced CAB-LSTM model was created and tested.

The experimental findings indicate that, the model proposed can predict the test set with an accuracy of 86.03% and Macro F1-Score of higher value than Logistic Regression, XGBoost, and a standard LSTM. It is recognized that there are two main avenues of research in the future:

A. Re-evaluation with Chronological Rigor

The most urgent would be to switch to a stringent chronological walk-forward validation approach that is used instead of random-split used presently. Walk-forward validation avoids the look-ahead bias and is the standard for consideration of time-series predictive models when used in real deployment scenarios. This re-assessment is required to get conservative and trustworthy real world performance estimates.

B. Architectural Enhancement via FinBERT

Although still useful, VADER-based sentiment analysis also works with a fixed vocabulary, and fails to recognize extensive semantics or local financial lingo. The following generation of the architecture proposed should swap the VADER with a domain-directed transformer like FinBERT [17, 18, 20] which is pre-trained on financial corpora, and recognizes wisely more rich semantic representations. The improvement is likely to create further increases in predictive accuracy, especially in the areas of intricate macroeconomic announcements and disclosures of earnings.

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